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Thapar Institute of Engineering and Technology, Patiala

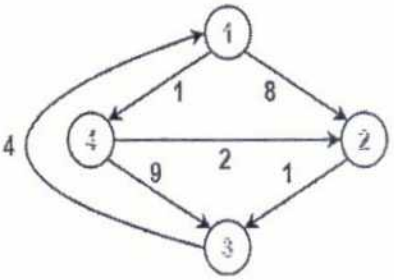
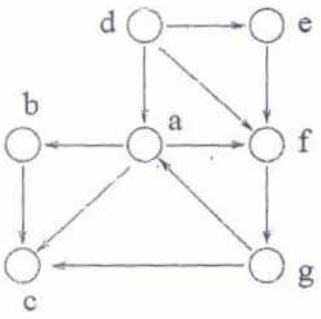
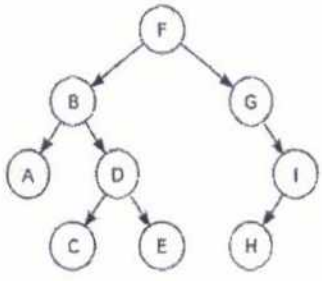
Department of Computer Science and Engineering

BE CSBS *(2nd Semester) EST* — UCT203: Data Structures and Algorithms

21st May 2023 — Time: 03Hours; MM: 80

Name of Faculty: Dr. Rinkle Rani

Instruction: Attempt all questions in sequence.

Q 1.	a) Solve All Pairs Shortest Path problem (Floyd's Algorithm) for the directed weighted graph given in figure 1. b) Define binary tree? Discuss the various memory representations techniques used for binary trees with the help of suitable example.	(10) (6)
Q 2.	Design a max heap from the following sequence of nodes. Apply Heap Sort algorithm on this heap. Draw all the intermediate trees for insertions and deletions. 38 24 44 21 55 47 67 49 What is hashing? Explain various hash functions used for hashing with the help of examples.	(12) (4)
Q 3.	a) Find all the nodes approachable from **d** of the graph given in figure 2 using BFS graph traversal algorithm. Show all the intermediate steps. b) For the binary tree given in figure 3 answer the following (i) Write inorder, preorder and postorder traversals. (ii) How much memory is required to store this binary tree sequentially?	(10) (6)
Q 4.	a) Design an extended binary tree using Hoffman Algorithm from the following sequence of nodes and find out the minimum weighted path length. 22 53 26 61 27 36 23 25 24 28 44 b) Develop a BST by inserting nodes from the following sequence one by one. 62 29 22 58 37 35 47 42 52 75 92 82 97 Perform the following operations on the above resulting tree independently. (i) Node 29 is deleted (ii) Node 75 is deleted (iii) Node 58 is deleted	(10) (6)
Q 5.	a) Develop an AVL tree from the following nodes, by inserting nodes one by one. Show all the intermediate steps and rotations. 8 9 10 2 1 5 3 6 4 7 11 12 b) Define Graph. Discuss the various memory representations techniques used for graph with the help of an example.	(12) (4)
	 figure 1  figure 2  figure 3	